

Fetal health risk prediction using CTG recordings

Abstract

Monitoring fetal health during pregnancy is crucial for ensuring the safety of both the mother and the unborn child. Cardiotocography (CTG) is a widely used medical technique that records fetal heart rate and uterine contractions to evaluate fetal well-being. However, interpreting CTG data manually can be time-consuming and may sometimes lead to subjective decisions. Therefore, there is a growing need for intelligent systems that can assist healthcare professionals in accurately analyzing fetal health data and identifying potential risks at an early stage.

The objective of this project is to develop a data-driven approach to classify fetal health conditions using machine learning techniques. The study utilizes the Fetal Health dataset, which contains several features derived from CTG measurements such as baseline fetal heart rate, accelerations, fetal movements, uterine contractions, and other statistical indicators. Based on these features, fetal health is categorized into three classes: Normal, Suspect, and Pathological. The goal of the system is to analyze these features and automatically predict the fetal health condition.

In this work, the dataset is first preprocessed by analyzing feature distributions, detecting potential outliers using the Interquartile Range (IQR) method, and normalizing the data using feature scaling techniques. Exploratory data analysis is performed using correlation analysis and visualization to understand the relationships between different features and fetal health outcomes. The dataset is then divided into training and testing sets to build predictive models.

Two supervised machine learning algorithms - Logistic Regression and k-Nearest Neighbors (k-NN) are implemented to classify fetal health conditions. These models are trained on the processed dataset and evaluated using standard performance metrics such as accuracy, precision, recall, and F1-score. A confusion matrix is also used to analyze the classification results and identify correctly and incorrectly predicted cases. The model with the best performance is selected as the final predictive model.

In addition to classification, unsupervised learning using the K-Means clustering algorithm is applied to explore natural groupings within the data and visualize patterns among fetal health indicators. The results demonstrate that machine learning techniques can effectively identify patterns in CTG data and assist in predicting fetal health conditions.

This project highlights the potential of data science and machine learning in supporting medical decision-making systems. By automating the analysis of fetal health data, the proposed approach can help healthcare professionals detect abnormalities early and improve prenatal care.